

Knowledge Soundness of Polynomial Commitments in the Algebraic Group Model Does Not Guarantee Extractability

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Abstract. The Algebraic Group Model (AGM) has become a standard framework for analyzing the knowledge soundness of group-based polynomial commitment schemes. In this work, we formally establish inherent limitations of this methodology. We isolate a structural property satisfied by essentially all practical group-based polynomial commitments, which we term AGM-clarity. We prove that for AGM-clear schemes, evaluation binding implies knowledge soundness in the AGM. This collapse reveals that the AGM definition of knowledge soundness does not capture a distinct security property, but is merely a structural consequence of evaluation binding.

To precisely characterize the guarantees on extractability provided by the AGM, we introduce Weak Interpolation Knowledge Soundness (WIKS) in the standard model, which is an extreme relaxation of extractability. We show that WIKS is implied by standard evaluation binding and prove that, for AGM-clear schemes, knowledge soundness in the AGM is equivalent to WIKS. Ultimately, our results demonstrate that proofs in the AGM for practical polynomial commitment schemes do not certify “knowledge” in the sense of immediate extractability, as they yield no guarantees beyond those already implied by standard model evaluation binding.

Keywords: Polynomial Commitment Schemes · Extractability · Knowledge Soundness · Evaluation Binding · Algebraic Group Model · AGM.

1 Introduction

Polynomial commitment schemes (PCS) are a central building block in modern zk-SNARKs, offering succinct openings and fast verification independent of the committed polynomial’s degree. While pairing-based [KZG10, PST13] and inner-product-based [BBB⁺18] constructions clearly win in proof size, yielding proofs of a few hundred bytes compared to the polylogarithmic overhead of post-quantum alternatives, their security proofs present significant theoretical challenges. Specifically, proving *knowledge soundness* for these group-based schemes in the standard model is highly non-trivial, culminating in the recent proofs under new, albeit falsifiable, hardness assumptions [LPS24, BDH⁺25].

To bypass these hurdles, security of polynomial commitments is frequently analyzed in the *Algebraic Group Model (AGM)* [FKL18]. In the AGM, the adversary is restricted to being *algebraic*: for every group element it outputs, it must provide an *algebraic explanation*, i.e., the coefficients of a linear combination of its inputs corresponding to the output. This restriction simplifies proofs dramatically. Gabizon, Williamson, and Ciobotaru [GWC19] noted that the AGM allows modeling binding and knowledge soundness in a single clean game, effectively permitting the extractor to return a polynomial immediately after the commitment (Remark 3.2 in [GWC19]). Indeed, for practical schemes, the algebraic explanation of a commitment *is* the coefficient vector of the polynomial, making extraction syntactic and the proof of knowledge soundness a straightforward reduction to a suitable q -DLog assumption. However, this simplicity raises a fundamental question:

If extraction is syntactic, what exactly is guaranteed by knowledge soundness in the AGM?

Prior Critiques of the AGM. Existing critiques of the AGM have focused on its *limited expressivity*. Lipmaa, Parisella, and Siim [LPS23] argue that the AGM forbids *oblivious sampling* of group elements without knowing their linear representations, which is a capability real adversaries possess, e.g., via admissible encodings in elliptic curves. They introduced the *AGM with Oblivious Sampling (AGMOS)* to remedy this, motivated by the observation that assumptions like SpurKE (the spurious knowledge assumption of immediate knowledge of exponent) are false in reality despite being provable in the AGM. While clearly valuable, this line of work critiques the *model*. Our motivation is different: we accept the AGM as a model but study the guarantees provided by the *definition* of knowledge soundness within it.

1.1 Our Results

In this work, we present formal arguments establishing intrinsic limitations of the notion of knowledge soundness for polynomial commitments in the AGM for a broad class of schemes encompassing all known group-based constructions previously analyzed in the AGM.

AGM-Clarity. We first define a structural property satisfied by existing group-based polynomial commitments, which we term *AGM-clarity*. A polynomial commitment scheme is AGM-clear if, given the algebraic explanation of a commitment C , one can efficiently construct a polynomial f consistent with C . As we discussed, this property is immediate for practical proposals such as Bulletproofs [BBB⁺18], KZG [KZG10], its multivariate variants [PST13], KZG-based Gemini [BCHO22], Zeromorph [KT24], Samaritan [GPS25] or Mercury [EG25] since, in these constructions, the commitment is a linear function of the group elements given to the adversary as the commitment key. Intuitively, AGM-clarity allows extracting a polynomial consistent with the commitment directly from its algebraic explanation provided by any AGM adversary. However, AGM-clarity is not a security property, and it alone does not suffice for knowledge soundness in the AGM. Specifically, even after extracting the polynomial f from the algebraic explanation of the commitment C , one has to argue the consistency of f with the subsequent evaluation claim (α, η) for which the adversary provides an accepting evaluation proof.

Evaluation Binding implies Knowledge Soundness in the AGM. Our initial observation is that, for AGM-clear schemes, evaluation binding already implies knowledge soundness. The extractor can simply output the polynomial f reconstructed from the adversary’s algebraic explanation of C . To argue that f is consistent with a claimed evaluation (α, η) for which the adversary produced an accepting proof π , suppose instead that $f(\alpha) \neq \eta$. Then one could use f with C to construct another accepting proof for $(\alpha, f(\alpha))$, contradicting evaluation binding.

Theorem 1 (Informal). *Every AGM-clear polynomial commitment scheme that is evaluation binding is also knowledge-sound in the AGM.*

Note that the conclusion of Theorem 1 is quite unexpected, as we show that a presumably strong security guarantee is a structural consequence of an arguably mild one. For comparison, in the standard model, evaluation binding is comparatively easy to establish under mild assumptions, e.g., for KZG under SBDH [KZG10], since its violations are local and admit straightforward reductions. By contrast, knowledge soundness demands extracting a *globally* consistent polynomial, a far stronger guarantee. In particular, proving the knowledge soundness of the KZG polynomial commitment in the standard model under assumptions sufficient for evaluation binding would likely require a significant conceptual breakthrough.

On extraction from the evaluation proof in the AGM. One could consider a relaxed notion of *extractability in the AGM* where the extractor, besides the commitment and its algebraic explanation, may also use the evaluation proof together with its algebraic explanation (cf. Definition 6 in [LPS23]). However, this relaxed notion is implied by the knowledge soundness in the AGM: an extractor that uses only the commitment with its algebraic explanation clearly succeeds also when given strictly more information. From Theorem 1, it then follows that, for AGM-clear schemes, evaluation binding implies also extractability in the AGM. In

other words, for practical group-based schemes, giving the extractor access to the evaluation proof does not significantly alter the security notion relative to evaluation binding, and the same collapse applies.

A Standard Model Analogue of Knowledge Soundness in the AGM. The above results suggest that, for polynomial commitments, the current formulation of knowledge soundness in the AGM provides a significantly weaker guarantee compared to knowledge soundness in the standard model. We formalize this intuition by defining a clearly restricted notion of knowledge soundness in the standard model, which we term *Weak Interpolation Knowledge Soundness (WIKS)*, and show that it is equivalent to knowledge soundness in the AGM for AGM-clear polynomial commitments. Thus, WIKS provides a tight upper bound on the standard model security guaranteed by the knowledge soundness in the AGM for AGM-clear PCS.

We define WIKS via a game between an adversary and an extractor that proceeds as follows: The adversary chooses a commitment C , and the extractor then asks $d + 1$ evaluation queries. To all of the queries, the adversary must provide values with accepting evaluation proofs, and, moreover, interpolating over these evaluation queries must yield a polynomial consistent with the commitment C . The adversary also outputs an evaluation point α , a claimed evaluation η and an accepting evaluation proof. Given the $d + 1$ answers to its evaluation queries with proofs, the extractor outputs a polynomial f of degree d , and the adversary wins iff $f(\alpha) \neq \eta$.

WIKS can be thought of as an extreme relaxation of special soundness for polynomial commitments. In the special soundness game (cf. Def. 5 in [LPS24]), the extractor would be given $d + 1$ accepting evaluation proofs for distinct evaluation points. However, there would be no guarantee on the consistency of the interpolating polynomial with the commitment C . Indeed, the crux of the standard model proof of special soundness for the univariate, respectively multivariate, KZG scheme (Lipmaa et al. [LPS24], respectively Belohorec et al. [BDH⁺25]) is showing that the interpolating polynomial is consistent with the commitment under a suitable falsifiable assumption.

Evaluation binding implies WIKS in the standard model. The main similarity between knowledge soundness in the AGM and WIKS is that, for AGM-clear schemes, there exists a canonical extractor: for knowledge soundness in the AGM, it follows from AGM-clarity, and, for WIKS, it is enforced by the requirement that the adversary must give evaluations that interpolate to a polynomial consistent with the originally given commitment. The existence of a canonical interpolating extractor allows us to prove the analogue of Theorem 1 also for WIKS, i.e., showing that WIKS collapses to evaluation binding in the standard model.

Theorem 2 (Informal). *Every polynomial commitment scheme that is evaluation binding is also weak interpolation-knowledge-sound.*

Under AGM-clarity, WIKS characterizes knowledge soundness in the AGM. Finally, we show that knowledge soundness in the AGM is characterized by WIKS in the standard model.

Theorem 3 (Informal). *Every AGM-clear polynomial commitment scheme is weakly interpolation-knowledge-sound if and only if it is knowledge-sound in the AGM.*

A depiction of the established relations is provided in Figure 1. In particular, for an AGM-clear polynomial commitment scheme Π , one can prove the knowledge soundness in the AGM by simply relying on any proof of evaluation binding for Π in the standard model.

The two directions of Theorem 3 are both proved via a simple *translation layer* that turns extractors into adversaries and vice versa while preserving their joint output distributions. The common ingredient is a *canonical reconstruction* of a polynomial: given $d+1$ accepting openings at distinct points, the extractor interpolates a degree- d polynomial φ and then checks that φ is consistent with the commitment C . AGM-clarity is used to turn φ into the algebraic explanation demanded by the AGM.

WIKS $\Rightarrow$ knowledge soundness in the AGM (Lemma 1): Fix any WIKS extractor Ext and wrap it into an AGM extractor Ext' : on input a commitment C , Ext' queries the adversary on $d+1$ fresh points, interpolates φ , and feeds to Ext the commitment together with the algebraic explanation induced by φ . If some

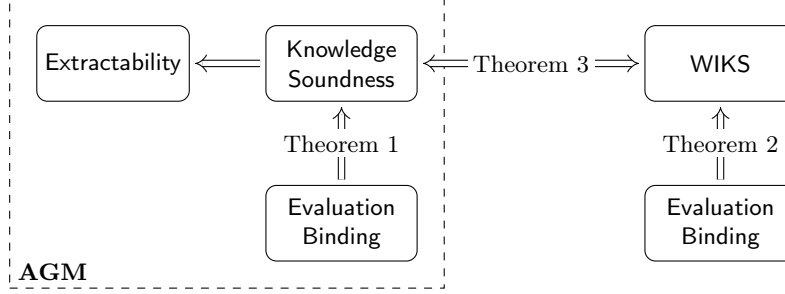


Fig. 1. Visual summary of the relationships established in this work. The dashed box highlights notions in the Algebraic Group Model. Theorems 1 and 3 assume additionally AGM-clarity of the PCS. Note that evaluation binding in the standard model implies evaluation binding in the AGM trivially.

AGM adversary $\mathcal{A}'$ breaks knowledge soundness against *every* AGM extractor, then, in particular, it breaks Ext'. Composing $\mathcal{A}'$ with the inverse translation yields a WIKS adversary that defeats the original Ext, contradicting WIKS.

Knowledge soundness in the AGM $\Rightarrow$ WIKS (Lemma 2): Starting from a WIKS adversary $\mathcal{A}'$ that defeats every WIKS extractor, we build an AGM adversary $\mathcal{A}$ against any fixed AGM extractor Ext. The adversary $\mathcal{A}$ emulates $\mathcal{A}'$ and, whenever Ext needs an algebraic explanation of C , it first obtains $d+1$ accepting openings from $\mathcal{A}'$ at distinct random points, interpolates φ , and (by AGM-clarity) supplies the corresponding algebraic explanation for C . Because the translation preserves the joint distributions of outputs of WIKS/AGM adversary and extractor, $\mathcal{A}$ inherits $\mathcal{A}'$'s non-negligible success, contradicting AGM knowledge soundness.

On Randomized/Multivariate Commitments and AGMOS. Our results in the following sections are stated and proven for randomized polynomial commitments, and they also address the common hiding/zero-knowledge variants of group-based polynomial commitments. They can be simplified for deterministic commitments simply by ignoring the randomness in the corresponding definitions. While our formal exposition treats univariate polynomial commitments for clarity, the extensions to multivariate commitments are straightforward and require no new ideas beyond routine notation (e.g., component-wise explanations and multivariate interpolation).

Our results establish relationships between notions in the AGM and the standard model. Belohorec, Hubáček, and Stejskal [BHS26] recently showed that, for many natural security games, AGM security can be lifted to the AGM with Oblivious Sampling (AGMOS) of Lipmaa et al. [LPS23] under the Find Polynomial Representation (FPR) and Tensor Oracle Find Representation (TOFR) assumptions. Among other applications, Belohorec et al. illustrated their general lifting theorem on knowledge soundness of polynomial commitments, and showed that knowledge soundness in the AGM implies knowledge soundness in the AGMOS for many KZG-like pairing-based polynomial commitments under these assumptions. Therefore under FPR and TOFR,³ our results can be lifted to the AGMOS, establishing that knowledge soundness in the AGMOS is also implied by evaluation binding and equivalent to WIKS. In other words, even moving to the AGMOS does not resolve the limitations of knowledge soundness in the AGM.

1.2 Summary and Open Problems

Given that the very premise of the AGM is to restrict adversaries to algebraic behavior, making witness extraction part of the model, one might wonder:

³ Essentially all non-trivial proofs of security in the AGMOS, and, in particular, of knowledge soundness of polynomial commitments (e.g., the direct proof of extractability for KZG in [LPS23]), rely on the TOFR assumption (and FPR, which reduces to PDL). Thus, we do not see them as significantly restrictive additional assumptions.

Is it so surprising that extractability comes almost for free in a model whose premise is extractability?

However, the above question is orthogonal to our motivation and our formal results. Our objective is not to critique the inherent properties of the AGM, but to clarify what the model actually guarantees for polynomial commitments. We have established that for AGM-clear schemes, knowledge soundness collapses into a structural corollary of evaluation binding. Therefore, AGM proofs for these schemes are effectively vacuous regarding “knowledge”: they do not certify immediate extractability, but rather our weaker notion of WIKS, which is already implied by standard evaluation binding.

These findings have two immediate implications. First, our community must be clear about the implications of proofs of security for polynomial commitments in the AGM. Practitioners may interpret knowledge soundness in the AGM as a guarantee of real-world extractability, unaware that for practical schemes, the proof adds no security assurances beyond those provided by evaluation binding. Second, since extractability in the AGM is largely an artifact of the model, genuine knowledge guarantees must be established in the standard model. Progress has been made in this direction using new falsifiable assumptions like ARSDH [LPS24, BDH⁺25, LPS25, CGKY25], but proving standard-model knowledge soundness under more classical assumptions remains a major open problem.

Regarding the scope of our results, removing the restriction to AGM-clear schemes is an open question. However, we view this restriction as *benign*: practically all group-based PCS are inherently AGM-clear due to the linearity of their commitment algorithms. Thus, our results cover essentially all schemes of practical interest.

1.3 Additional Related Work

Until recently, the knowledge soundness of group-based polynomial commitments was established either under non-falsifiable assumptions [ZGK⁺17, CHM⁺20] or in idealized models such as the AGM [GWC19, BDFG20, BDFG21, GW21, GPS25, EG25]. The drawbacks of non-falsifiable assumptions have long been recognized [Nao03].⁴ Regarding idealized models, the knowledge soundness of group-based polynomial commitments has been analyzed primarily in the Algebraic Group Model (AGM) of Fuchsbauer, Kiltz, and Loss [FKL18]. As the AGM is a relatively recent framework, its limitations and relationship to other idealized models like the Generic Group Model (GGM) [Sho97, Mau05] remain the subject of ongoing research [KP19, AHK20, ABK⁺21, ZZK22, AHPP23, JM24, BFHK24].

Various works leverage AGM to analyze the knowledge soundness and related notions, even for higher-level protocols. [BMM⁺21] and [GMN22] used AGM to prove soundness of batch Groth16 proofs. [GT21] defined state-restoration soundness in the AGM, and gave a proof of state-restoration soundness in the AGM for Bulletproofs. The Halo2 book claims state-restoration soundness in the AGM for Halo2 [Com21]. Note that state-restoration soundness is stronger than knowledge soundness. Finally, [ZBK⁺22, PK22, EFG22, ZGK⁺25] considered lookup arguments and analysed their security in the AGM.

2 Preliminaries

We recall the key notions, starting with polynomial commitment schemes. For conciseness, we consider the group parameter generation to be a part of the definition of a PCS.

Definition 1 (PCS). *A non-interactive univariate polynomial commitment scheme (PCS) ($\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf}$) consists of the following polynomial-time algorithms:*

Setup: $\text{PGen}(1^\lambda) \rightarrow \text{gp}$; *generate group parameters.*

Commitment key generation: $\text{KGen}(\text{gp}, d) \rightarrow \text{ck}$; *given group parameters and a degree bound d , return a commitment key ck , which implicitly contains gp .*

⁴ A practical downside is that relying on non-falsifiable assumptions, such as Power-Knowledge-of-Exponent, effectively doubles the scheme’s complexity, as parties must execute two “parallel” instances of the protocol.

Commitment: $\text{Com}(\text{ck}, f(X); r) \rightarrow C$; for a polynomial $f(X) \in \mathbb{F}[X]^{\leq d}$, compute a (possibly randomized) commitment C .

Opening: $\text{Open}(\text{ck}, C, \alpha, f(X); r) \rightarrow (\eta, \pi)$; to open a commitment C to a polynomial $f(X) \in \mathbb{F}[X]^{\leq d}$ at a point $\alpha \in \mathbb{F}$, return the evaluation $\eta = f(\alpha)$ and an evaluation proof π .

Verification: $\text{Vrf}(\text{ck}, C, \alpha, \eta, \pi) \rightarrow \{0, 1\}$; intuitively, accept (1) or reject (0) the proof π that η is the evaluation at the point α of some “promised” polynomial $f(X) \in \mathbb{F}[X]^{\leq d}$ behind the commitment C .

The above definition also covers randomised schemes such as the “blinded” KZG scheme (see, e.g., Section 3.5.3 in Kohrita and Towa [KT24]), where the commitment and opening functions take some additional randomness r to achieve hiding.

As the goal of a polynomial commitment scheme is to force the prover to commit to a polynomial, it should (among other requirements) be able to ensure that the prover cannot make a commitment which can be opened to multiple distinct values at the same evaluation point. This is formalized as the evaluation binding property of a PCS.

Definition 2 (Evaluation binding). A polynomial commitment scheme $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ is evaluation binding if for any $d \in \text{poly}(\lambda)$ and PPT adversary $\mathcal{A}$,

$$\Pr \left[\begin{array}{l} \text{Vrf}(\text{ck}, C, \alpha, \eta, \pi) = 1 \wedge \\ \text{Vrf}(\text{ck}, C, \alpha, \eta', \pi') = 1 \wedge \eta \neq \eta' \end{array} \middle| \begin{array}{l} \text{gp} \leftarrow \text{PGen}(1^\lambda); \text{ck} \leftarrow \text{KGen}(\text{gp}, d); \\ (C, \alpha, \eta, \pi, \eta', \pi') \leftarrow \mathcal{A}(\text{ck}) \end{array} \right] \in \text{negl}(\lambda).$$

In words, the probability that $\mathcal{A}$ wins the following evaluation-binding game is negligible. Given a commitment key ck , $\mathcal{A}$ must open some commitment C at some evaluation point α to two different purported evaluations η, η' so that both proofs π, π' are accepted.

We continue with introducing the Algebraic Group Model and algebraic adversaries in particular.

Definition 3 (AGM). Let PGen be a group parameter generator. A PPT algorithm $\mathcal{A}$ is algebraic (or an AGM adversary) if, for any group elements it outputs, it also outputs their algebraic explanations with respect to its input. Formally, for any $\text{gp} \leftarrow \text{PGen}(1^\lambda)$ and a vector of input group elements $\bar{g}$,

$$\mathcal{A}(\text{gp}, \bar{g}) \rightarrow (\bar{o}, \gamma_{\bar{o}}),$$

where $\gamma_{\bar{o}}$, the algebraic explanation of $\mathcal{A}$ ’s output $\bar{o}$, is a matrix over our field $\mathbb{F}$, and it holds that

$$\bar{o} = \gamma_{\bar{o}} \bar{g}. \tag{1}$$

The input or output of $\mathcal{A}$ may contain some other field elements, which we omit. In the Algebraic Group Model (AGM), all PPT adversaries are algebraic.

The crucial property of a PCS that is of interest to us is knowledge soundness in the AGM, which informally ensures both the property of evaluation binding and a requirement on the prover to “know” the polynomial it commits itself to.

Definition 4 (Knowledge Soundness in the AGM). A polynomial commitment scheme $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ is knowledge-sound in the AGM if, for any $d \in \text{poly}(\lambda)$, there exists a PPT extractor Ext such that for every algebraic adversary $\mathcal{A}$,

$$\Pr \left[\begin{array}{l} \text{Vrf}(\text{ck}, C, \alpha, \eta, \pi) = 1 \wedge \\ (C \neq \text{Com}(\text{ck}, f(X); r) \vee \deg(f(X)) > d \vee f(\alpha) \neq \eta) \end{array} \middle| \begin{array}{l} \text{gp} \leftarrow \text{PGen}(1^\lambda); \text{ck} \leftarrow \text{KGen}(\text{gp}, d); \\ ((C, \gamma_C), \alpha, \eta, \pi) \leftarrow \mathcal{A}(\text{ck}); \\ (f(X), r) \leftarrow \text{Ext}(\text{ck}, (C, \gamma_C)) \end{array} \right] \in \text{negl}(\lambda),$$

where γ_C is the algebraic explanation of C .

The above definition is adapted from the definitions of knowledge-soundness in the AGM from the literature (see, e.g., [GWC19, BDFG20, BDFG21, LPS23]). Note that we do not require $\mathcal{A}$ to explain the group elements in the evaluation proof π since the extractor has access only to the algebraic explanation of the commitment anyway. It is natural to consider a relaxation of knowledge soundness in the AGM that only requires extraction after seeing the evaluation proof with its algebraic explanation, as formalized below. Trivially, a scheme that is knowledge sound is also extractable.

Definition 5 (Extractability in the AGM). A polynomial commitment scheme $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ is extractable in the AGM if, for any $d \in \text{poly}(\lambda)$ and every algebraic adversary $\mathcal{A}$, there exists a PPT extractor $\text{Ext}_{\mathcal{A}}$ such that

$$\Pr \left[\begin{array}{l} \text{Vrf}(\text{ck}, C, \alpha, \eta, \pi) = 1 \wedge \\ (C \neq \text{Com}(\text{ck}, f(X); r) \vee \deg(f(X)) > d \vee f(\alpha) \neq \eta) \end{array} \middle| \begin{array}{l} \text{gp} \leftarrow \text{PGen}(1^\lambda); \text{ck} \leftarrow \text{KGen}(\text{gp}, d); \\ ((C, \gamma_C), \alpha, \eta, (\pi, \gamma_\pi)) \leftarrow \mathcal{A}(\text{ck}); \\ (f(X), r) \leftarrow \text{Ext}(\text{ck}, (C, \gamma_C), \alpha, \eta, (\pi, \gamma_\pi)) \end{array} \right] \in \text{negl}(\lambda),$$

where γ_C and γ_π are the algebraic explanations of C and π , respectively.

3 Evaluation Binding Implies Knowledge Soundness in the AGM

In this section, we show that, in the AGM, evaluation binding implies knowledge soundness for a large class of polynomial commitments. We start with defining the polynomial commitments to which our results apply.

Definition 6 (AGM-clarity). A PCS $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ is AGM-clear if there exists a DPT revealer Rev such that, for every $d \in \text{poly}(\lambda)$, the following implication holds. If $\text{gp} \leftarrow \text{PGen}(1^\lambda)$, $\text{ck} \leftarrow \text{KGen}(\text{gp}, d)$, C is a commitment and γ_C its algebraic explanation with respect to ck , then $(f(X), r) \leftarrow \text{Rev}(\text{ck}, (C, \gamma_C))$, where $\deg(f(X)) \leq d$ and $\text{Com}(\text{ck}, f(X); r) = C$.

Informally, given the algebraic explanation of a commitment, the revealer allows efficient extraction of a polynomial consistent with the commitment. We note that many polynomial commitments from the literature, such as Bulletproofs [BBB⁺18], KZG [KZG10], its multivariate variants [PST13], Gemini [BCHO22], Zeromorph [KT24], Samaritan [GPS25] or Mercury [EG25], as well as their randomized hiding variants, are all AGM-clear with a revealer that simply outputs the algebraic explanation of the commitment.

Theorem 1. Let $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ be an AGM-clear PCS. If Π is evaluation binding, then it is knowledge-sound in the AGM.

Proof. We prove the statement by establishing the contrapositive. Let Π be an AGM-clear PCS that is not knowledge-sound in the AGM. We show that then Π is not evaluation binding.

To break the evaluation binding of Π , we leverage any adversary $\mathcal{A}$ that wins the knowledge-soundness game against the most basic extractor Ext^* that simply outputs the polynomial produced from commitment C and its algebraic explanation γ_C by the revealer Rev guaranteed by AGM-clarity of Π . Since Π is not knowledge-sound in the AGM, for any efficient extractor Ext , there exists an algebraic adversary $\mathcal{A}$ such that it can win the knowledge-soundness game against Ext for Π with non-negligible probability. In particular, there exists such a successful adversary against the above revealer-based extractor Ext^* . Thus, given the commitment key ck , the adversary $\mathcal{A}$ can, with non-negligible probability, output a commitment C with an algebraic explanation γ_C and a point α with a claimed evaluation η and an accepting proof π such that the polynomial that $\text{Ext}^*(\text{ck}, (C, \gamma_C))$ outputs evaluates to a value $\eta' \neq \eta$ and $\text{Vrf}(\text{ck}, C, \alpha, \eta, \pi) = 1$.

Given a commitment key ck , our adversary $\mathcal{A}'$ for the evaluation binding game proceeds as follows:

1. Run $\mathcal{A}(\text{ck})$ to get a commitment C with an explanation γ_C , and a point α with a claimed evaluation η and a proof π such that $\text{Vrf}(\text{ck}, C, \alpha, \eta, \pi) = 1$.

2. Run $\text{Rev}(\text{ck}, (C, \gamma_C))$ to obtain a polynomial ψ and randomness r such that $\text{Com}(\text{ck}, \psi(X); r) = C$. Compute $\eta' = \psi(\alpha)$ and the corresponding evaluation proof $\pi' = \text{Open}(\text{ck}, C, \alpha, \psi(X); r)$.
3. Output $(C, \alpha, \eta, \eta', \pi, \pi')$.

The above adversary $\mathcal{A}'$ wins the evaluation-binding game with the same probability of $\mathcal{A}$ winning the AGM knowledge-soundness game against the extractor Ext^* , which we assume to happen with non-negligible probability. Hence, $\mathcal{A}'$ wins the evaluation binding game with non-negligible probability, establishing the contrapositive, and, thus, the theorem. $\square$

4 Translating Knowledge Soundness in the AGM to the Standard Model

In this section, we characterize the power of knowledge soundness in the AGM (Definition 4) with respect to the standard model. To this end, we define a standard model notion of *weak interpolation knowledge soundness* (WIKS) and then show that WIKS is equivalent to knowledge soundness in the AGM for all AGM-clear polynomial commitment schemes.

In the definition of WIKS below, we use Lagrange interpolation, which we recall for completeness. For two vectors $\bar{x}, \bar{\eta} \in \mathbb{F}^d$, we use

$$\text{Interpolate}(X, \bar{x}, \bar{\eta}) := \sum_{i=1}^d \eta_i \prod_{j=1, j \neq i}^d \frac{X - x_j}{x_j - x_i}$$

to denote the degree d Lagrange interpolation polynomial in variable X such that the value at point x_i is η_i .

Definition 7 (Weak interpolation knowledge soundness in the standard model). For a PCS $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$, a degree bound $d \in \text{poly}(\lambda)$, a PPT extractor $\text{Ext} = (\text{Ext}_1, \text{Ext}_2)$, and an adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, the weak interpolation knowledge soundness experiment $\text{WIKS}_{\mathcal{A}, \text{Ext}}^{\Pi, d}(\lambda)$ is defined as follows:

1. $\text{gp} \leftarrow \text{PGen}(1^\lambda); \text{ck} \leftarrow \text{KGen}(\text{gp}, d)$
2. $(C, \text{state}) \leftarrow \mathcal{A}_1(\text{ck})$
3. $(x_1, \dots, x_{d+1}) \leftarrow \text{Ext}_1(\text{ck}, C)$
4. $((\eta_1, \pi_1), \dots, (\eta_{d+1}, \pi_{d+1})), (\alpha, \eta, \pi), r \leftarrow \mathcal{A}_2(\text{state}, \text{ck}, C, x_1, \dots, x_{d+1})$
5. $f(X) \leftarrow \text{Ext}_2(\text{ck}, C, ((x_1, \eta_1, \pi_1), \dots, (x_{d+1}, \eta_{d+1}, \pi_{d+1})), r)$
6. If all of the following hold, return 1, otherwise, return 0:
 - (a) $\text{Vrf}(\text{ck}, C, \alpha, \eta, \pi) = 1$
 - (b) for all $i \in [d+1]$, $\text{Vrf}(\text{ck}, C, x_i, \eta_i, \pi_i) = 1$
 - (c) for $\bar{x} = (x_1, \dots, x_{d+1})$ and $\bar{\eta} = (\eta_1, \dots, \eta_{d+1})$,

$$C = \text{Com}(\text{ck}, \text{Interpolate}(X, \bar{x}, \bar{\eta}); r)$$

- (d) $C \neq \text{Com}(\text{ck}, f(X); r) \vee \deg(f(X)) > d \vee f(\alpha) \neq \eta$

We say that a polynomial commitment scheme Π satisfies weak interpolation knowledge soundness if there exists a PPT extractor $\text{Ext} = (\text{Ext}_1, \text{Ext}_2)$ such that, for all adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ and degree bounds $d \in \text{poly}(\lambda)$, it holds that $\Pr [\text{WIKS}_{\mathcal{A}, \text{Ext}}^{\Pi, d}(\lambda) = 1] \in \text{negl}(\lambda)$.

In words, the adversary chooses a commitment C , the extractor then asks $d+1$ evaluation queries. To all of the queries, the adversary must provide values with accepting evaluation proofs, and randomness r such that interpolating over these evaluation queries must yield a polynomial consistent with the commitment C under the randomness r . The adversary also outputs an evaluation point α , a claimed evaluation η and an accepting evaluation proof. Given the $d+1$ answers to its evaluation queries with proofs, the extractor outputs a polynomial f of degree d consistent with C under the randomness r , and the adversary wins iff $f(\alpha) \neq \eta$.

Note that WIKS is, as its name suggests, indeed quite limiting as it effectively requires the adversary to have some globally consistent strategy for winning the game. In particular, there are clearly many standard-model adversaries outside the scope of WIKS we must consider when studying the full-blown notion of knowledge soundness in the standard model.

WIKS vs. Special Soundness. Lipmaa et al. [LPS24] introduced a seemingly similar standard model notion of *special soundness* for polynomial commitments. In their special soundness experiment (Definition 5 in [LPS24]), the evaluation points $(x_1, \dots, x_{d+1})$ are chosen by the adversary (under the condition of being pairwise distinct to ensure a valid interpolation domain), and, crucially, the winning condition does not contain item 6.(c), i.e., the adversary can, in principle, provide evaluation points and values that do not interpolate to a polynomial consistent with the commitment C . Since there is no syntactic guarantee on the consistency of the provided evaluations with the commitment, one proves the consistency of the extracted polynomial with the commitment under computational assumptions (ARSDH in [LPS24]). On the other hand, item 6.(c) in the winning condition in WIKS ensures syntactically that extraction via interpolation is successful, making WIKS a strictly weaker notion.

Evaluation binding implies WIKS in the standard model. First, we show that, for any PCS, an analogue of Theorem 1 holds for WIKS as well. This is due to the fact that, for WIKS, there exists a canonical extractor in the vein of the revealer for AGM-clear polynomial commitments. Indeed, to succeed in the WIKS experiment, the adversary must provide point evaluations that interpolate to a polynomial consistent with the originally output commitment.

Theorem 2. *Let $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ be a PCS. If Π is evaluation binding, then it is weakly interpolation-knowledge-sound.*

Proof. We prove this using the contrapositive: the fact that weak interpolation knowledge soundness is broken gives us an adversary breaking evaluation binding. Let us consider the following extractor $\text{Ext}' = (\text{Ext}'_1, \text{Ext}'_2)$ for weak interpolation knowledge soundness: we query the adversary in $d + 1$ distinct uniformly random points, and from these evaluations, we interpolate a polynomial. Then, by the fact that weak interpolation knowledge soundness is broken, we get an adversary $\mathcal{A}' = (\mathcal{A}'_1, \mathcal{A}'_2)$ that wins the weak interpolation knowledge soundness game with non-negligible probability.

Using $\mathcal{A}' = (\mathcal{A}'_1, \mathcal{A}'_2)$, we construct an adversary $\mathcal{A}$ for evaluation binding as follows. We take the commitment C generated by $\mathcal{A}'_1$, and ask $\mathcal{A}'_2$ for $d + 1$ correct evaluation proofs of the committed polynomial in distinct uniformly random points. Then, we interpolate the polynomial φ using the received evaluations. By our assumptions on $\mathcal{A}'$, we know that φ can be committed to C . Moreover, as $\mathcal{A}'$ breaks weak interpolation knowledge soundness, it also outputs a point α with claimed evaluation η and a proof π that correctly verifies such that $\varphi(\alpha) \neq \eta$ with non-negligible probability.

Therefore, with non-negligible probability, we have two distinct evaluations of the committed polynomial at point α : η with the proof π , and $\varphi(\alpha)$, where the proof can be generated by our knowledge of the polynomial. Thus, evaluation binding is also broken. $\square$

In fact, for the previous proposition, we can also weaken the assumptions on the adversary slightly by letting the extractor run in expected polynomial time and allowing the adversary to fail by outputting evaluation proofs which do not pass verification. In that case, to have any chance of winning with non-negligible probability, the evaluation proofs of the adversary must be correct with non-negligible probability. In particular, this implies that in expectation, we will need $\mathcal{O}(d \cdot \lambda^c)$ queries, which would still amount to a polynomial number of steps.

Knowledge soundness in the AGM vs. WIKS in the standard model. Next, we establish the equivalence.

Theorem 3. *Let $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ be an AGM-clear PCS. Then Π satisfies WIKS if and only if it is knowledge-sound in the AGM.*

The proof of Theorem 3 is established via the two corresponding reductions presented in Lemmas 1 and 2 below. Our strategy in constructing both reductions uses a “translation layer” between an adversary and an extractor such that it can be joined with either to form a larger adversary or extractor depending on our current need, depicted in Figures 2 and 3.

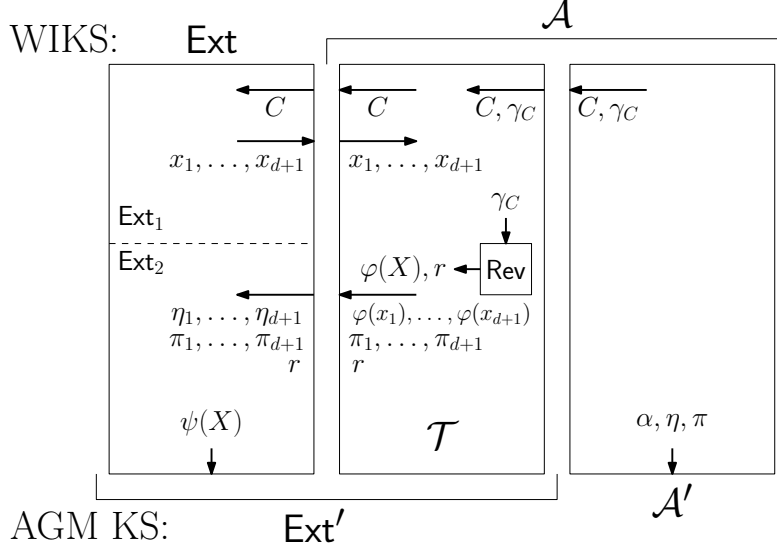


Fig. 2. A visual representation of the reduction establishing Lemma 1.

Lemma 1. *Let $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ be an AGM-clear PCS. If Π satisfies WIKS then it is knowledge-sound in the AGM.*

Proof. Again, we prove this using the contrapositive: if knowledge soundness in the AGM is broken, then so is weak interpolation knowledge soundness. As Π is AGM-clear, let Rev be the revealer certifying its AGM-clarity.

Let $\text{Ext} = (\text{Ext}_1, \text{Ext}_2)$ be a WIKS extractor for which we want to construct an adversary that breaks it. We modify Ext into an extractor Ext' for knowledge soundness in the AGM by composing Ext with a translation layer $\mathcal{T}$ as depicted in Figure 2: when the algebraic adversary outputs a commitment C with an algebraic explanation γ_C , we use the revealer Rev to transform γ_C into a description of a polynomial φ and randomness r consistent with C , which we then use to truthfully answer the queries asked by Ext_1 , i.e., for all $1 \leq i \leq d+1$, set $\eta_i = \varphi(x_i)$ and $\pi_i = \text{Open}(\text{ck}, C, x_i, \eta_i; r)$. The polynomial ψ that is the final output of Ext_2 is then returned unchanged.

Since Ext' is an extractor for knowledge soundness in the AGM, there exists an adversary $\mathcal{A}'$ that wins the knowledge-soundness game against Ext' with non-negligible probability. From $\mathcal{A}'$, we now construct the adversary $\mathcal{A}$ against the original extractor Ext . On a high-level, we take $\mathcal{A}'$ and add the translation layer $\mathcal{T}$ we used to modify Ext into Ext' . Formally, we modify $\mathcal{A}'$ into $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ as follows: in $\mathcal{A}_1$, the commitment C with its algebraic explanation γ_C output by $\mathcal{A}'$ is translated using Rev into a description of a polynomial φ and randomness r consistent with the commitment C . The adversary $\mathcal{A}_1$ sets $\text{state} = (\varphi, r)$ and outputs (C, state) . Then, the evaluation queries of Ext_1 are answered truthfully by $\mathcal{A}_2$ using φ and r from state as above. When $\mathcal{A}'$ outputs an evaluation point α , the claimed evaluation η , and a purported proof π , $\mathcal{A}_2$ returns them unchanged.

By construction, the joint distributions of outputs of $\mathcal{A}$ and Ext , respectively $\mathcal{A}'$ and Ext' , are identical. Moreover, by the definition of knowledge soundness in the AGM, if $\mathcal{A}'$ wins against Ext' then $\psi(\alpha) \neq \eta$. Thus, for the same joint output, $\mathcal{A}$ wins against Ext in the WIKS game. Thus, $\mathcal{A}$ wins the WIKS game against Ext with at least the probability of $\mathcal{A}'$ winning the AGM knowledge soundness game against Ext' , which we assumed to be non-negligible. $\square$

Lemma 2. *Let $\Pi = (\text{PGen}, \text{KGen}, \text{Com}, \text{Open}, \text{Vrf})$ be an AGM-clear PCS. If Π is knowledge-sound in the AGM then it satisfies WIKS.*

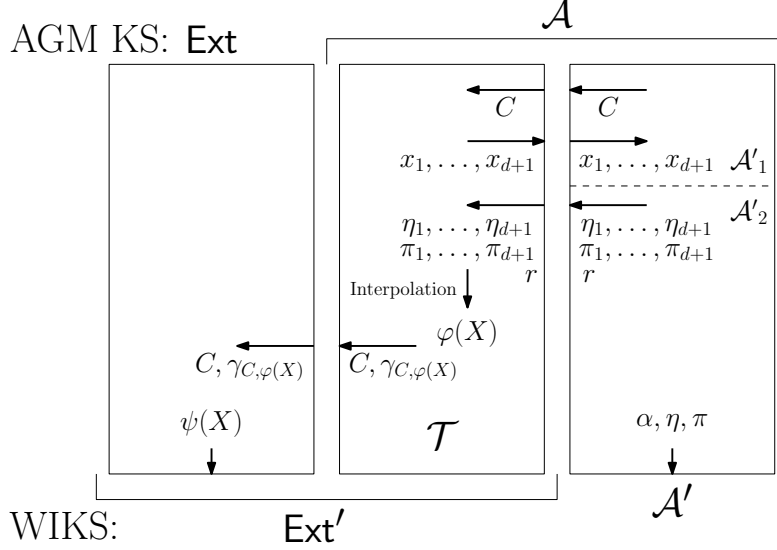


Fig. 3. A visual representation of the reduction establishing Lemma 2.

Proof. As before, we prove the contrapositive: WIKS being broken implies broken knowledge soundness in the AGM.

Given an extractor Ext for knowledge soundness in the AGM, we construct an algebraic adversary $\mathcal{A}$ that breaks it. We construct an extractor $\text{Ext}' = (\text{Ext}'_1, \text{Ext}'_2)$ for WIKS from the original extractor Ext using the translation layer $\mathcal{T}$ depicted in Figure 3. First, Ext'_1 takes the provided commitment C and generates distinct uniformly random evaluation points to query the WIKS adversary $\mathcal{A}'$. Ext'_2 then interpolates a polynomial φ through the evaluations received from the adversary, which is guaranteed to be consistent with C under the provided randomness r (conditioned on $\mathcal{A}'$ winning). It then passes to Ext the commitment C with the algebraic explanation γ_C obtained when computing C as $\text{Com}(\text{ck}, \varphi; r)$.⁵ When Ext outputs the final polynomial ψ , Ext'_2 returns ψ as well.

As Ext' is an extractor for WIKS, there exists an adversary $\mathcal{A}' = (\mathcal{A}'_1, \mathcal{A}'_2)$ that wins the WIKS game with non-negligible probability. We now construct the algebraic adversary $\mathcal{A}$ similarly as in the previous lemma: starting with $(\mathcal{A}'_1, \mathcal{A}'_2)$, we use the commitment C given by $\mathcal{A}'_1$, and generate distinct uniformly random points that are given to $\mathcal{A}'_2$. Using the evaluations from $\mathcal{A}'_2$, we interpolate a polynomial φ consistent with the commitment C and use φ to create the algebraic explanation for C . The evaluation point α , the claimed value η , and proof π output by $\mathcal{A}'_2$ are returned unchanged.

By construction, the joint distributions of outputs of $\mathcal{A}$ and Ext , respectively $\mathcal{A}'$ and Ext' , are identical. Moreover, by the definition of WIKS, if $\mathcal{A}'$ wins against Ext' then $\psi(\alpha) \neq \eta$. (By Item 6(c) in the definition of WIKS, we are guaranteed correctness of the interpolation, and thus the first two disjuncts of Item 6(d) cannot be true, which ensures that $\psi(\alpha) \neq \eta$.) Thus, for the same joint output, $\mathcal{A}$ wins against Ext in the AGM knowledge soundness game. Thus, $\mathcal{A}$ wins the AGM knowledge soundness game against Ext with at least the probability of $\mathcal{A}'$ winning the WIKS game against Ext' , which we assumed to be non-negligible. $\square$

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⁵ Here, we implicitly assume that the commitment algorithm Com is algebraic, which is without loss of generality. Specifically, the PCS Π is assumed to be knowledge sound in the AGM, and it would not make sense to restrict adversaries to being algebraic unless the algorithms of Π are also algebraic.

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